Explainable Artificial Intelligence (XAI) Mini Project (30 Marks)
Rubrics and Guidelines for Review

1. Objective

The objective of this mini project is to enable students to apply the concepts learned in Explainable Artificial Intelligence (XAI) by developing an interpretable machine learning application.

The project should integrate concepts from:

- Problem Identification
- Data Collection & Preprocessing
- Machine Learning Model Development
- Explainable AI Techniques
- Model Evaluation
- Visualization & Interpretation
- Final Demonstration

2. Project Theme

Example Domains

- Healthcare Disease Prediction
- Credit Risk Assessment
- Loan Approval Prediction
- Customer Churn Prediction
- Employee Attrition Analysis
- Fake News Detection
- Sentiment Analysis
- Crop Yield Prediction
- Air Quality Prediction
- Traffic Accident Prediction
- Fraud Detection
- Intrusion Detection System
- Sepsis Prediction
- Diabetes Prediction
- Heart Disease Prediction
- Cancer Diagnosis
- Student Performance Prediction
- Stock Price Prediction
- Energy Consumption Forecasting
- Insurance Claim Prediction

3. Project Development Phases

1. Problem Identification & Dataset Collection
2. Data Preprocessing & Feature Engineering
3. Machine Learning Model Development
4. Explainability Analysis using XAI Techniques
5. Visualization & Interpretation
6. Performance Evaluation
7. Final Demonstration

4. Report Preparation

The report should be 25–30 pages.

Formatting:

- Times New Roman
- 12 pt body, 25 pt cover title, 16–18 pt headings
- Single spacing
- Justified alignment
- Normal margins
- Page numbering

Suggested Contents:

1. Abstract
2. Introduction
3. Literature Review
4. Dataset Description
5. Data Preprocessing
6. Machine Learning Model
7. Explainable AI Methodology
8. Experimental Results
9. Visualization and Interpretation
10. Conclusion
11. References

5. Evaluation

Criterion	Marks	Excellent	Good	Needs Improvement
Review-1 (10 Marks)				
Problem Identification & Dataset Selection	3	Novel and Well-defined real-world problem with appropriate dataset	Problem mostly clear with suitable dataset	Problem statement or dataset unclear/inappropriate

Data Preprocessing & Feature Engineering	4	Complete preprocessing with feature engineering and justification	Minor preprocessing issues	Incomplete preprocessing
Exploratory Data Analysis (EDA)	3	Comprehensive visualizations and insights	Adequate analysis	Limited or incorrect analysis
Review-2 (10 Marks)				
Machine Learning Model Development	3	Appropriate model selection, training and tuning	Minor implementation issues	Incorrect/incomplete model
Explainable AI Implementation	4	Correct implementation of SHAP/LIME/Grad-CAM with explanations	Most methods implemented	Limited or incorrect explainability
Performance Evaluation	3	Comprehensive metrics and comparison with existing approaches	Adequate evaluation	Limited evaluation
Review-3 (10 Marks)				
Interpretation, Visualization & Deployment	4	Clear interpretation with effective visualization	Minor issues	Poor interpretation
Testing, Demonstration & Viva	6	Complete demo, report, viva	Adequate demonstration	Weak demonstration/documentation

Review Dates:

Review 1 Date: 21.08.26

Review 2 Date: 23.09.26

Review 3 Date: 21.10.26